

ARCHITECTURE SELECTION AND JUSTIFICATION

Blood Bank Inventory & Emergency Donor Matcher

Architecture Selection

We chose Layered Architecture for the Blood Bank Inventory & Emergency Donor Matcher System.

Architectural Choice

The system is organized into three layers: Presentation, Business and Data. The Presentation Layer handles interactions for emergency requesters and blood bank managers. The Business Layer handles emergency request processing, blood inventory management, donor matching, shelf-life monitoring and notifications. The Data Layer stores blood inventory, donor and emergency request information.

Reason 1

Layered Architecture provides clear separation of concerns. Emergency requests, inventory management, donor matching and data storage are separated into appropriate layers, making the system easier to understand, maintain and modify.

Reason 2

The system has clearly defined operations such as checking real-time blood inventory, monitoring blood component shelf lives, matching eligible donors and triggering emergency notifications. A layered structure provides a straightforward organization for these related functions without introducing the operational complexity of Microservices Architecture.

Security Advantage

The layered structure can restrict direct access to sensitive inventory, donor and emergency-request data. Users interact through the Presentation Layer, while the Business Layer controls access to the Data Layer, reducing unnecessary direct database access.

Performance Benefit

The Business Layer can process emergency matching requests and retrieve only the required inventory and donor information from the Data Layer. This separation keeps presentation activities lightweight and supports responsive emergency matching and notification.